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THE EFFECT OF THE USE OF ARTIFICIAL INTELLIGENCE BY MANAGERS ON THE OPERATIONAL INDICATORS OF IT COMPANIES: ANALYSIS OF THE LARGEST RUSSIAN FIRMS

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AI USAGE DECLARATION

Sections where AI tools were applied	Purpose of using artificial intelligence	Method of application	Tool name	Link to the tool	Evaluation
Introduction	To give the author's text an academic style	Author's text with the use of generated materials	ChatGPT	https://chatgpt.com/	The tool supplemented the basic understanding of the IT industry (5/5)
Literature Review	Clarification of wording, structuring of information, and giving the author's text an academic style	Author's text with the use of generated materials	ChatGPT	https://chatgpt.com/	In some cases, corrections were required to the information provided by AI (3/5)
Methods and Data	Clarification of wording, giving the author's text an academic style, and correction of grammatical errors	Author's text with the use of generated materials	ChatGPT	https://chatgpt.com/	The tool did not always correctly project the wording specified by the authors (3/5)
Results	Giving the author's text an academic style and	Author's text with the use of generated materials	ChatGPT	https://chatgpt.com/	No significant distortion of the author's text was observed (4/5)

	clarification of wording				
Discussion	Clarification of wording, giving the author's text an academic style, and correction of grammatical errors	Author's text with the use of generated materials	ChatGPT	https://chatgpt.com/	No significant distortion of the author's text was observed (4/5)
Conclusion	Clarification of wording, structuring of information, and giving the author's text an academic style	Author's text with the use of generated materials	ChatGPT	https://chatgpt.com/	No significant distortion of the author's text was observed (4/5)

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Abstract

This study examines the relationship between managerial AI integration and the operational indicators of large Russian IT companies, as well as the moderating role of firm size in this relationship. Prior research shows that AI adoption and AI-related capabilities may improve firm performance; however, less is known about whether managers' use of AI in operational decision-making is associated with measurable operational outcomes. Drawing on the resource-based view, dynamic capabilities theory, and contingency theory, this study conceptualizes AI as an organizational capability whose value depends on complementary resources, managerial routines, and organizational context.

The empirical analysis is based on manually collected panel data for 23 large Russian IT companies over the period 2019–2025. The dependent variables are labor productivity, return on assets, and capital expenditure intensity. Managerial AI integration is used as the key independent variable, while firm size is introduced as a moderating variable. The hypotheses are tested using regression models with heteroskedasticity-robust standard errors and year effects.

The results show that managerial AI integration is not positively and significantly associated with any of the three operational indicators. Therefore, the hypothesis about the direct positive relationship between managers' AI use and operational indicators is not supported. However, firm size positively moderates the relationship between managerial AI integration and ROA, while no significant moderation effect is found for labor productivity or capital expenditure intensity.

These findings suggest that managerial AI integration does not automatically translate into operational improvements. Instead, larger firms may have more favorable organizational conditions under which managerial AI integration is associated with profitability-related outcomes. The study contributes to AI and management research by shifting attention from general AI adoption to managerial AI integration and by providing evidence from the underexplored context of large Russian IT companies.

Keywords: managerial AI integration, operational performance, firm size, Russian IT companies, resource-based view, dynamic capabilities.

Introduction

1.1. Research background and motivation

Artificial intelligence (AI) has become an important source of organizational transformation and competitive advantage. In management and information systems research, AI is increasingly analyzed not only as a technological tool, but also as an organizational capability. Mikalef and Gupta (2021) define AI capability as the firm's ability to select, orchestrate, and leverage AI-specific resources. This view shifts attention from the mere presence of AI technologies to the way firms combine AI with data, human expertise, organizational routines, and managerial processes.

Prior research shows that AI and related data-driven capabilities may improve firm outcomes, but these effects are not automatic. Mikalef et al. (2020) show that big data analytics capability affects competitive performance indirectly through dynamic and operational capabilities, while Dubey et al. (2020) demonstrate that big data analytics powered by AI is associated with operational performance under specific organizational and environmental conditions. Therefore, AI-related technologies create value when they are embedded into organizational capabilities and used to support managerial and operational processes.

Despite the growing literature on AI and firm performance, several limitations remain. First, existing studies primarily examine AI adoption or AI capability at the firm level, while the direct role of managers' use of AI in operational decision-making remains less developed. Second, prior research often focuses on aggregate firm performance, productivity, or innovation, whereas specific operational indicators are less frequently examined together. Third, most empirical evidence comes from developed economies or broad cross-industry samples, while technology-intensive sectors in emerging-market contexts remain underexplored.

Large Russian IT companies provide a relevant setting for addressing these gaps. They operate in a knowledge-intensive environment where operational outcomes depend on managers' ability to process information, coordinate complex activities, and make timely decisions. The Russian context is also important because Molodchik et al. (2024) show that ICT profiles of Russian companies are heterogeneous and that the relationship between ICT combinations and financial performance varies across industries. Therefore, it remains an open empirical question whether AI-performance relationships documented in other contexts also hold for large Russian IT companies.

1.2. Research problem

The research problem addressed in this study is that the relationship between managers' use of AI and firms' operational performance remains insufficiently understood. Prior research often examines AI adoption as a general technological event, while less attention is paid to AI use as a managerial and organizational process. As a result, it remains unclear whether the integration of AI into managerial decision-making is associated with measurable operational outcomes.

This problem is especially relevant because AI implementation may involve organizational barriers and adjustment costs. Tariq et al. (2021) identify cultural constraints, fear of the unknown, lack of employee skills, and insufficient strategic planning as barriers to achieving operational excellence through AI. Cannas et al. (2024) show that AI implementation in operations and supply chain management may also be constrained by data quality problems, lack of specific skills, high investment requirements, unclear economic benefits, and limited experience in cost analysis for AI projects. Therefore, the presence of AI in corporate communication does not necessarily mean that AI is deeply embedded into operational decision-making.

The effect of managerial AI integration may also depend on firm size. Larger firms may possess greater data volumes, more developed digital infrastructure, stronger financial resources, and more formalized managerial systems. Babina et al. (2024) show that AI-related growth is concentrated among larger firms, while Ardito et al. (2025) demonstrate that AI adoption is more beneficial when combined with complementary digital technologies such as IoT and big data analytics. For this reason, firm size is considered not only as a control variable, but also as a moderating condition.

1.3. Relevance of the research problem

This research problem is theoretically relevant because it shifts attention from AI adoption to AI use as a managerial process. From the resource-based view, AI can be treated as a valuable organizational capability only when it is combined with complementary resources and embedded into firm-specific routines. Mikalef and Gupta (2021) support this logic by showing that AI capability emerges from the orchestration of AI-specific resources rather than from technology alone.

The problem is also relevant from the perspective of dynamic capabilities theory. Mikalef et al. (2020) show that data-based capabilities create performance value through dynamic capabilities,

which then influence operational capabilities. This means that AI-supported insights may affect firm outcomes only when organizations are able to reconfigure resources, routines, and decision-making processes.

The practical relevance of the problem lies in the need to understand whether AI-supported decision-making is associated with measurable operational outcomes. This issue is important for large Russian IT companies, which operate under external uncertainty and need to maintain productivity, profitability, and investment activity. If AI is not connected to concrete managerial routines and operational processes, it may remain a symbolic initiative rather than a source of measurable performance improvement.

1.4. Research questions

Accordingly, this study addresses two research questions.

RQ1: How does the use of artificial intelligence by managers relate to the operational indicators of large Russian IT companies?

RQ2: How does firm size moderate the relationship between managerial AI integration and the operational indicators of large Russian IT companies?

The study is grounded in the resource-based view, dynamic capabilities theory, and contingency theory. These theoretical lenses explain AI as an organizational capability whose value depends on complementary resources, resource reconfiguration, and alignment between technological systems, managerial capabilities, and organizational processes. This logic also suggests that firm size may condition the relationship between managerial AI integration and operational indicators.

1.5. Methods overview

Empirically, the study applies a quantitative research design based on panel data for 23 large Russian IT companies over the period from 2019 to 2025. The initial dataset includes 161 firm-year observations. After data cleaning, the final analytical sample for the regression models includes 149 observations. The dependent variables are labor productivity, return on assets, and capital expenditure intensity. Managerial AI integration is used as the key independent variable, while firm size is introduced as the moderating variable. The empirical analysis is based on regression models with heteroskedasticity-robust standard errors and year effects.

1.6. Preview of the main results

The results show that managerial AI integration is not positively and significantly associated with labor productivity, ROA, or capital expenditure intensity. Thus, the first hypothesis is not supported. The moderation analysis provides partial support for the second hypothesis: firm size does not significantly moderate the relationship between managerial AI integration and labor productivity or capital expenditure intensity, but it positively and significantly moderates the relationship with ROA. These findings suggest that managerial AI integration is associated with profitability-related outcomes more favorably in larger firms, rather than with immediate improvements in labor productivity or investment intensity.

1.7. Proposed practical implications and intended scientific contribution

This study contributes to the literature in three ways. First, it shifts the focus from AI adoption and AI capability to managers' use of AI in operational decision-making. Second, it examines several operational indicators simultaneously and shows that the relationship between managerial AI integration and firm outcomes differs across productivity, profitability, and investment intensity. Third, it extends empirical evidence to large Russian IT companies, a context that remains underrepresented in prior AI and management research.

Practically, the study suggests that managers should not expect AI use to automatically improve short-term productivity, profitability, or capital expenditure intensity. Instead, AI-supported decision-making may produce measurable outcomes only when it is supported by complementary organizational resources and embedded into managerial routines. The findings also indicate that firm size may strengthen the relationship between managerial AI integration and profitability-related outcomes, but this effect is not universal across all operational indicators.

1.8. Structure of the paper

The remainder of the paper is structured as follows. Section 2 reviews the literature on AI as an organizational capability, AI and operational performance, and firm size as a resource condition for AI value creation. Section 3 describes the data, variables, and empirical methodology. Section 4 presents the empirical results. Section 5 discusses the findings in relation to prior literature. Section 6 concludes the paper by summarizing the main findings, theoretical contribution, practical implications, limitations, and directions for future research.

2.Literature review

2.1. Artificial intelligence as an organizational capability

Artificial intelligence is increasingly considered not only as a technological tool, but also as an organizational capability. Mikalef and Gupta (2021) define AI capability as the ability of a firm to select, orchestrate, and leverage AI-specific resources. This view is relevant for the present study because the value of AI depends not on the mere presence of algorithms, but on how firms combine AI with data, human expertise, organizational routines, and managerial processes.

The resource-based logic explains why AI may become a source of performance differences between firms. Mikalef and Gupta (2021) show that AI capability is built through tangible resources, human skills, and intangible organizational resources, and that it is positively associated with organizational creativity and firm performance. Thus, AI creates value through capability building rather than through technological adoption alone.

Prior studies also show that the performance effects of AI-related technologies are often indirect. Wamba-Taguimdje et al. (2020) demonstrate that AI capabilities improve organizational performance through process-oriented dynamic capabilities. Mikalef et al. (2020) similarly show that big data analytics capability affects competitive performance through dynamic and operational capabilities. These findings suggest that AI-related technologies influence firm outcomes when analytical insights are transformed into managerial actions and operational routines.

This logic is important for defining managerial AI integration. In this study, AI is not treated only as an adopted technology. It is analyzed through its connection with managerial decision-making, planning, monitoring, and resource allocation. Therefore, managerial AI integration reflects the extent to which AI-related information is embedded into organizational processes.

2.2. Artificial intelligence and operational performance

Prior research links AI-related capabilities with operational performance through information processing, forecasting, resource allocation, and process improvement. Dubey et al. (2020) show that big data analytics powered by AI is positively associated with operational performance when it is aligned with organizational capabilities and environmental conditions. For data-intensive firms, this mechanism is important because operational decisions depend on timely and structured information.

AI may also improve operational performance when it is embedded into specific business processes. Cannas et al. (2024) show that AI applications in supply chain and operations management support planning, sourcing, production, delivery, and customer-related activities. Their study also indicates that AI can help firms reduce costs and lead times and improve service levels, quality, safety, and sustainability. Thus, AI contributes to operational performance when it is applied to concrete managerial and operational tasks.

At the same time, AI-related operational improvements are not automatic. Tariq et al. (2021) identify cultural constraints, fear of the unknown, lack of employee skills, and insufficient strategic planning as barriers to achieving operational excellence through AI. Cannas et al. (2024) also emphasize barriers related to data quality, lack of AI-specific skills, high investments, unclear economic benefits, and limited experience in cost analysis. These barriers explain why AI adoption or AI-related communication may not immediately produce measurable operational improvements.

Other studies examine AI in relation to broader firm-level outcomes. Mikalef and Gupta (2021) find that AI capability is positively associated with firm performance. Babina et al. (2024) show that firms investing in AI experience higher sales growth, employment growth, and market valuation, mainly through product innovation. Ardito et al. (2025) show that AI adoption is positively associated with revenue growth and that this effect becomes stronger when AI is combined with IoT and big data analytics. These findings support the expectation that AI can improve firm outcomes, but they also show that the effect depends on complementary resources and implementation context.

The present study treats operational performance as a multidimensional outcome. Labor productivity, ROA, and capital expenditure intensity capture different aspects of firm operations. Since prior research links AI to productivity, profitability, innovation, and process-level improvements through different mechanisms, managerial AI integration may affect these indicators differently. Nevertheless, the reviewed literature provides theoretical grounds to expect a positive relationship between managerial AI integration and operational indicators.

H1. The use of artificial intelligence by managers positively relates to the operational indicators of large Russian IT companies.

2.3. Firm size as a resource condition for AI value creation

Prior research suggests that the performance effects of AI depend on complementary organizational resources. Mikalef and Gupta (2021) argue that AI capability is created through the combination of tangible resources, human skills, and intangible organizational resources. Therefore, AI value creation depends not only on technology, but also on the resources that support its use.

Firm size may influence the extent to which companies benefit from managerial AI integration. Larger firms usually have broader data resources, stronger financial capacity, more developed digital infrastructure, and more formalized managerial systems. These conditions may help firms embed AI-generated insights into planning, monitoring, resource allocation, and operational coordination.

Empirical evidence supports this logic. Babina et al. (2024) show that AI-related growth is concentrated among larger firms and is associated with higher sales, employment, and market valuation. Ardito et al. (2025) show that AI adoption is more beneficial when combined with complementary digital technologies such as IoT and big data analytics. These findings indicate that AI generates stronger outcomes when firms possess the resources needed to scale and integrate AI across business processes.

The Russian context strengthens the relevance of this argument. Molodchik et al. (2024) show that ICT profiles of Russian companies differ across industries and that the relationship between ICT combinations and financial performance depends on specific combinations of technologies and resources. This supports the view that AI-related value depends on organizational context and resource complementarity.

2.4. Contingency theory as a theoretical foundation for the moderating role of firm size

Contingency theory provides a theoretical foundation for explaining why the relationship between managerial AI integration and operational indicators may depend on firm size. The theory rejects the idea of universally effective managerial practices and argues that organizational effectiveness depends on the fit between managerial tools, internal organizational characteristics, and contextual conditions (Woodward, 1965). In the context of AI, this means that the same level of managerial AI integration may produce different outcomes depending on the resources and organizational capacity of the firm.

This logic is consistent with empirical studies on AI and performance. Dubey et al. (2020) show that big data analytics powered by AI improves operational performance under the influence of organizational and environmental conditions. Babina et al. (2024) find that AI-related growth is concentrated among larger firms, while Ardito et al. (2025) show that AI adoption is more beneficial when combined with complementary digital technologies. Together, these studies support the argument that AI creates stronger performance effects when firms possess the resources and complementary capabilities needed to use it effectively.

In this study, firm size is treated as a contingency factor. Larger firms may be better positioned to connect AI-supported decision-making with operational outcomes because they usually have stronger infrastructure, larger data resources, and more formalized managerial systems. Therefore, the relationship between managers' use of AI and operational indicators is expected to be stronger in larger firms.

H2. Firm size positively moderates the relationship between managerial AI integration and the operational indicators of large Russian IT companies, so that this relationship is stronger for larger firms.

2.5. Research gap and research model

The reviewed literature shows that AI can affect firm performance through organizational capabilities, operational processes, and complementary resources. Mikalef and Gupta (2021) conceptualize AI as a firm-level capability, while Mikalef et al. (2020) and Wamba-Taguimdje et al. (2020) show that the performance effects of data-driven and AI-related capabilities are mediated by dynamic and operational capabilities. However, three gaps remain important for the present study.

First, prior studies mainly examine AI adoption or AI capability at the firm level, while the role of managers' use of AI in operational decision-making is less developed. Second, existing research often focuses on aggregate firm performance, productivity, or innovation, whereas several operational indicators are less frequently examined together. Third, the Russian IT sector remains underexplored, although Molodchik et al. (2024) show that Russian companies differ in their ICT profiles and that the relationship between ICT combinations and financial performance varies across industries.

Accordingly, the proposed research model includes managerial AI integration as the independent variable; labor productivity, ROA, and capital expenditure intensity as dependent variables; and

firm size as the moderating variable. This framework allows the study to test whether managerial AI integration is associated with operational outcomes and whether this association becomes stronger in larger firms.

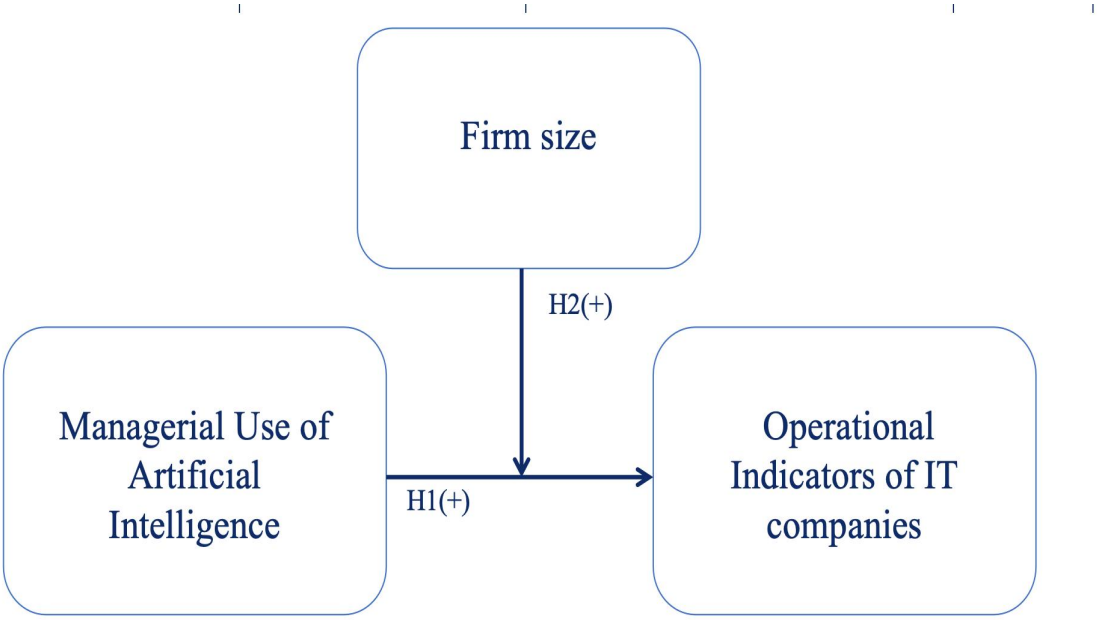


Figure 1. Theoretical model

3. Methods and Data

3.1 Research design

This study applies a quantitative empirical research design to examine the relationship between managerial AI integration and the operational indicators of large Russian IT companies. The empirical analysis is based on firm-level panel data covering the period from 2019 to 2025. This design corresponds to the objective of the study because it allows testing whether managerial AI integration is associated with operational outcomes and whether this relationship depends on firm size.

The empirical strategy directly follows the research questions and hypotheses. Hypothesis 1 examines the direct relationship between managerial AI integration and operational indicators. Hypothesis 2 examines whether firm size moderates this relationship. Accordingly, the study estimates baseline regression models for the direct effect and moderation models with an interaction term between managerial AI integration and firm size.

The analysis focuses on three operational indicators: labor productivity, return on assets, and capital expenditure intensity. These indicators reflect different dimensions of firm-level operational outcomes. Labor productivity captures the efficiency of labor use, return on assets reflects the efficiency of asset utilization in generating profit, and capital expenditure intensity represents investment-related operational activity. Since these indicators are conceptually different and not interchangeable, the regression models are estimated separately for each dependent variable.

3.2 Sample and data sources

The dataset was manually compiled from publicly available corporate and financial sources. The main sources include annual reports, financial statements prepared under Russian Accounting Standards or IFRS where available, official company websites, corporate press releases, and public disclosures related to AI initiatives and digital transformation. Publicly available industry rankings and corporate information were used to identify large Russian IT companies and verify their relevance to the research context.

The final dataset includes 23 large Russian IT companies observed over the period from 2019 to 2025. The initial dataset consisted of 161 firm-year observations. After data cleaning and the exclusion of observations with missing values in the variables required for regression analysis, the final analytical sample included 149 firm-year observations.

The selected period is relevant for the study because it captures the active diffusion of AI technologies and the growing role of AI in corporate management and operational decision-making. At the same time, the period includes major external shocks, including the COVID-19 pandemic, sanctions imposed on Russia from 2022, and the restructuring of the Russian IT market. Therefore, year effects are included in the empirical models to control for common time-specific shocks that could affect firm performance independently of AI use.

The use of public corporate sources is appropriate for this study because large Russian IT companies are more likely to disclose financial information, strategic initiatives, and technology-related activities. These sources make it possible to collect comparable firm-level information on financial indicators, firm characteristics, and AI-related activities. However, this approach also has limitations. Public disclosures may reflect not only actual internal AI use but also corporate communication strategies. Therefore, managerial AI integration is interpreted as a disclosure-based proxy for publicly observable AI-related managerial activity rather than as a direct measure of all internal managerial AI practices.

3.3 Sample construction and data filtering

The sample was constructed using a criterion-based sampling approach. Companies were included in the dataset if they met three conditions. First, the company had to operate in the Russian IT sector or have IT-related activities as a core part of its business model. Second, the company had to belong to the group of large market participants. This criterion was assessed using publicly available rankings of Russian IT companies, revenue-based lists, and corporate disclosures. In the final dataset, the scale criterion is reflected in the composition of the sample: it includes 23 companies with publicly available financial and operational data for 2019–2025, whose revenue in the latest observed year ranges from RUB 5 billion to RUB 4,089 billion. This range confirms that the selected companies represent sizeable market actors rather than small firms. Third, the company had to provide sufficient public information to construct the variables used in the empirical analysis.

The initial list of companies was checked manually. Firms were excluded if they did not provide comparable financial information, if their public disclosures did not allow the construction of AI-related variables, or if the available data were incomplete for the key variables required for regression analysis. As a result, the final sample represents a non-random but theoretically relevant group of large Russian IT companies.

The sample is not intended to represent the entire Russian IT sector. Many small and medium-sized IT companies operate as limited liability companies and do not publish detailed financial statements. As a result, their financial and operational data are often unavailable or not comparable across firms and years. The focus on large companies increases the reliability and comparability of the collected data but limits the generalizability of the findings to smaller and less transparent firms.

Before model estimation, the dataset was cleaned and prepared for analysis. Observations with missing values in variables required for a particular model were excluded. Duplicate firm-year observations were checked and removed where necessary. Visible data entry errors were corrected where possible. Extreme values in the dependent variables were examined to reduce the risk that regression estimates would be driven by abnormal observations.

3.4 Variables

The dependent variables capture three dimensions of operational performance.

The first dependent variable is labor productivity. It reflects the efficiency with which a firm converts labor resources into output. Labor productivity is calculated as revenue divided by the number of employees. Since the original variable is highly skewed due to substantial differences in firm scale, its logarithmic form is used in the regression analysis.

The second dependent variable is return on assets. ROA is used as a profitability-related operational indicator. It reflects how efficiently a company uses its asset base to generate profit and is calculated as net profit divided by total assets.

The third dependent variable is capital expenditure intensity. This variable captures investment-related operational activity and reflects the extent to which firms allocate resources to capital development. It is included because managerial AI integration may be associated not only with productivity and profitability, but also with investment behavior, technological upgrading, and modernization of operational processes.

The main independent variable is managerial AI integration. This variable reflects the degree to which AI is connected to managerial and organizational activities. It is calculated as the ratio of AI-related managerial mentions to total AI-related mentions:

$$managerial_ai_integration = \frac{ai_manager_mentions}{ai_mentions_total} \quad (1)$$

If the total number of AI mentions equals zero, managerial AI integration is coded as zero. The variable ranges from 0 to 1. Higher values indicate that a larger share of AI-related corporate communication is connected to managerial, organizational, or decision-making activities.

AI-related mentions were identified in corporate reports and public communications. A general AI mention refers to explicit references to artificial intelligence, machine learning, neural networks, natural language processing, computer vision, predictive analytics, or similar AI-related technologies. A managerial AI mention was coded when AI was explicitly connected to decision-making, planning, forecasting, monitoring, resource allocation, organizational coordination, management processes, risk assessment, or operational control. Therefore, the variable does not measure AI adoption in general, but specifically captures the relative presence of AI references linked to managerial and organizational functions.

Firm size is used both as a control variable and as the moderator in Hypothesis 2. It is measured as the natural logarithm of total assets. The logarithmic transformation is used because total assets differ substantially across companies, and the logarithmic form reduces the effect of extreme differences in scale. Firm size is theoretically important because larger firms may possess greater financial, organizational, and technological resources for implementing AI-based managerial practices.

The control variables include firm age and leverage. Firm age is measured as the number of years since the company's foundation. This variable is included because older firms may differ from younger firms in organizational routines, accumulated experience, and operational structure. Leverage is measured as total debt divided by total assets. This variable controls for the financial structure of firms, as debt burden may influence profitability, investment behavior, and operational performance.

For the moderation analysis, managerial AI integration and firm size are mean-centered before constructing the interaction term. The interaction term is calculated as follows:

$$ai_size_interaction = managerial_ai_c * log_assets_c \quad (2)$$

Centering is applied to reduce potential multicollinearity between the interaction term and its components and to make the interpretation of the main effects more stable.

3.5 Empirical strategy

The empirical analysis uses regression models to test the proposed hypotheses. The models are estimated separately for each operational indicator: log labor productivity, ROA, and capital expenditure intensity. Robust standard errors are used to improve the reliability of statistical inference.

To test Hypothesis 1, the following baseline specification is estimated:

$$Yit = \beta 0 + \beta 1\, Managerial_AI_Integrationit + \beta 2\, Firm_Sizeit + \beta 3\, Firm_Ageit + \\ + \beta 4\, Leverageit + \lambda t + \varepsilon it \quad (3)$$

where Yit denotes the operational indicator of firm i in year t. The dependent variable is measured alternatively as log labor productivity, ROA, or capital expenditure intensity. Managerial_AI_Integrationit is the main independent variable. Firm_Sizeit, Firm_Ageit, and Leverageit are control variables. λt represents year effects, and εit is the error term.

The coefficient β1 is the key parameter for testing Hypothesis 1. A positive and statistically significant coefficient would indicate that managerial AI integration is positively associated with the corresponding operational indicator.

To test Hypothesis 2, the following moderation specification is estimated:

$$Yit = \beta 0 + \beta 1\, Managerial_AI_cit + \beta 2\, Firm_Size_cit + \beta 3\, AI_Size_Interactionit + \\ + \beta 4\, Firm_Ageit + \beta 5\, Leverageit + \lambda t + \varepsilon it \quad (4)$$

In this model, Managerial_AI_cit is the centered managerial AI integration variable, Firm_Size_cit is the centered firm size variable, and AI_Size_Interactionit is their interaction term. The coefficient β3 is the key parameter for testing the moderating role of firm size. A positive and statistically significant coefficient of the interaction term would indicate that firm size strengthens the relationship between managerial AI integration and the corresponding operational indicator.

3.6 Model diagnostics and validity checks

Several diagnostic procedures are used to assess the validity of the empirical models.

First, descriptive statistics are calculated for all key variables. This step provides information about the mean, standard deviation, minimum, and maximum values of the dependent variables, independent variable, moderator, and controls.

Second, a Pearson correlation matrix is calculated to examine preliminary associations between the variables. The correlation matrix is used to assess whether the dependent variables capture different dimensions of operational performance and whether the explanatory variables are highly correlated with one another.

Third, multicollinearity is assessed using the Variance Inflation Factor. VIF diagnostics are conducted for both the baseline and moderation specifications. This is particularly important for the moderation models because the inclusion of an interaction term may increase the correlation between explanatory variables. The reported VIF values are below the conventional threshold of 5, indicating that multicollinearity does not represent a serious concern in the estimated models.

Fourth, year effects are included in all regression models to control for time-specific shocks. This is important because the period from 2019 to 2025 includes substantial external changes that may influence all firms simultaneously. Including year effects helps reduce the risk that the estimated associations are driven by common macroeconomic or institutional factors.

The results are interpreted as associations rather than causal effects. Although the study uses firm-level panel data, control variables, year effects, robust standard errors, and diagnostic tests, the observational nature of the dataset does not allow the full elimination of reverse causality or omitted variable bias.

All empirical analyses were conducted in R. The analysis included data cleaning, variable construction, descriptive statistics, correlation analysis, VIF diagnostics, and regression estimation.The R script used for the empirical analysis and the final dataset are provided as separate supplementary files attached to the final submission in EIOS.

3.7 Methodological limitations

Several methodological limitations should be acknowledged.

First, managerial AI integration is measured using publicly available corporate disclosures. This approach makes the variable transparent and replicable, but it may not fully capture the actual internal use of AI by managers. Companies may differ in their communication strategies, and firms that actively disclose AI-related initiatives may not necessarily use AI more intensively in internal managerial processes.

Second, the sample is limited by the availability of public financial and corporate information. Many Russian IT companies do not disclose detailed financial statements, especially if they operate as limited liability companies. Therefore, the sample is biased toward larger and more transparent firms.

Third, the study focuses on large Russian IT companies. This context is relevant for examining AI integration in a technology-intensive industry, but the results may not be generalizable to small firms, non-IT companies, or firms operating in other countries.

Fourth, although the models include control variables, year effects, and robust standard errors, unobserved firm characteristics may still influence both managerial AI integration and operational performance. Therefore, the estimated coefficients should not be interpreted as causal effects.

Finally, the period of analysis includes major external shocks that could influence firms' financial and operational indicators. Year effects partly address this issue, but they cannot fully capture all firm-specific responses to external changes.

Despite these limitations, the selected research design is suitable for the objective of the study. The dataset provides firm-level evidence on large Russian IT companies, the variables correspond to the theoretical framework, and the empirical strategy allows the study to test both the direct association between managerial AI integration and operational indicators and the moderating role of firm size.

4. Results

4.1 Preliminary Data Analysis

Table 1 presents the descriptive statistics for the variables included in the empirical analysis. The dependent variables capture three dimensions of operational performance: labor productivity, return on assets, and capital expenditure intensity. Labor productivity is measured in logarithmic form. Its mean value is 2.32, with a standard deviation of 1.14. The values range from -0.33 to 9.39, indicating substantial variation in productivity levels across the observed firms.

Return on assets has a mean value of 0.18 and a standard deviation of 0.38. The minimum value is -0.41, while the maximum value reaches 3.36. This indicates that the sample includes both firms with negative profitability and firms with relatively high profitability. Capital expenditure intensity has a mean value of 0.15 and a standard deviation of 0.13, with values ranging from 0.01 to 0.80.

The main independent variable, managerial AI integration, has a mean value of 0.19 and a standard deviation of 0.29. Since the variable ranges from 0 to 1, these values indicate substantial differences in the extent to which AI-related corporate communication is connected to managerial and organizational activities. The AI adoption variable has a mean value of 0.91. In categorical terms, 91.28% of the observations refer to firms with AI adoption, while 8.72% refer to firms without AI adoption.

The control variables also demonstrate variation across the sample. Firm size, measured as the logarithm of total assets, has a mean value of 11.56 and a standard deviation of 3.01. Firm age has a mean value of 22.33 years and ranges from 4 to 34 years. Leverage has a mean value of 0.43 and a standard deviation of 0.24, with values ranging from 0.00 to 1.35.

Overall, the descriptive statistics show that the sample is heterogeneous in terms of operational performance, AI-related characteristics, and firm-level controls. This variation provides the basis for the subsequent regression analysis.

Table 1. Descriptive statistics

Variable	Mean	SD	Minimum	Maximum	Category	Percent
Dependent variables						
Labor productivity (log)	2.32	1.14	-0.33	9.39		
ROA	0.18	0.38	-0.41	3.36		
Capex intensity	0.15	0.13	0.01	0.8		
Independent variables and moderator						
Managerial AI integration	0.19	0.29	0	1		
AI adoption	0.91	0.28	0	1	No AI adoption	8.72
					AI adoption	91.28
Controls						
Firm size (log assets)/Moderator	11.56	3.01	5.01	18.05		
Firm age	22.33	7.85	4	34		
Leverage	0.43	0.24	0	1.35		

Table 2 presents the Pearson correlation matrix for the main variables included in the empirical analysis. The dependent variables are not strongly correlated with one another, which supports their use as separate indicators of operational performance. Labor productivity is positively and significantly correlated with capital expenditure intensity ($r = 0.19$, $p < 0.01$). ROA is negatively and significantly correlated with capital expenditure intensity ($r = -0.14$, $p < 0.05$). The correlation between labor productivity and ROA is close to zero.

Managerial AI integration is negatively and significantly correlated with ROA ($r = -0.23$, $p < 0.001$). At the same time, managerial AI integration is not significantly correlated with labor productivity ($r = -0.08$) or capital expenditure intensity ($r = 0.05$). These results suggest that the relationship between managerial AI integration and operational indicators is not uniform across the three dependent variables.

The control variables also show several significant associations with the dependent variables. Firm size is positively and significantly correlated with labor productivity ($r = 0.21$, $p < 0.01$) and capital expenditure intensity ($r = 0.28$, $p < 0.001$). However, firm size is negatively and significantly correlated with ROA ($r = -0.50$, $p < 0.001$). Firm age is negatively and significantly

correlated with ROA ($r = -0.32$, $p < 0.001$), while its correlations with labor productivity and capital expenditure intensity are not statistically significant.

The correlation matrix also indicates that managerial AI integration is positively and significantly correlated with firm size ($r = 0.41$, $p < 0.001$). This means that larger firms tend to demonstrate higher levels of AI integration in managerial activities. At the same time, the reported correlations among explanatory variables do not indicate severe multicollinearity. Therefore, the variables can be included in the regression models.

Table 2. Pearson correlation matrix

Variables	1	2	3	4	5	6
1. Labor productivity	1					
2. ROA	-0.01	1				
3. Capital investment intensity	0.19**	-0.14*	1			
4. Managerial AI integration	-0.08	-0.23***	0.05	1		
5. Firm size (log assets)	0.21**	-0.5***	0.28***	0.41***	1	
6. Firm age	-0.1	-0.32***	-0.04	0.06	0.26***	1

*Note. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.*

4.2 Main analysis

Hypothesis 1: The Direct Effect of Managerial AI Integration

Hypothesis 1 states that managerial AI integration is positively associated with the operational indicators of large Russian IT companies. To test this hypothesis, three regression models were estimated separately for log labor productivity, ROA, and capital expenditure intensity. Managerial AI integration was included as the main independent variable. Firm size, firm age, and leverage were included as control variables, and year effects were added to all specifications. All models were estimated on the analytical sample of 149 firm-year observations.

Before interpreting the regression results, multicollinearity was assessed using the Variance Inflation Factor. The results of the VIF diagnostics are reported in Appendix A, Table A1. All reported VIF values are below the conventional threshold of 5, indicating that multicollinearity does not represent a concern in the models used to test Hypothesis 1.

The regression results are presented in Table 3. In Model 1, where log labor productivity is used as the dependent variable, managerial AI integration has a negative and statistically significant coefficient ($\beta = -0.91$, $p < 0.01$). This indicates that managerial AI integration is significantly associated with labor productivity, but the direction of the relationship is negative rather than positive. Therefore, Model 1 does not support Hypothesis 1.

Among the control variables in Model 1, firm size has a positive and statistically significant coefficient ($\beta = 0.18$, $p < 0.001$), while firm age has a negative and statistically significant coefficient ($\beta = -0.03$, $p < 0.001$). Leverage is not statistically significant. The adjusted R^2 of Model 1 is 0.21, indicating that the model explains 21% of the variation in log labor productivity. The model is statistically significant overall.

In Model 2, where ROA is used as the dependent variable, managerial AI integration has a negative but statistically insignificant coefficient ($\beta = -0.09$, $p > 0.05$). Therefore, the model does not provide evidence of a statistically significant relationship between managerial AI integration and ROA. Model 2 does not support Hypothesis 1.

Among the control variables in Model 2, firm size has a negative and statistically significant coefficient ($\beta = -0.06$, $p < 0.001$). Firm age and leverage are not statistically significant. The adjusted R^2 of Model 2 is 0.11, indicating that the model explains 11% of the variation in ROA. The model is statistically significant overall.

In Model 3, where capital expenditure intensity is used as the dependent variable, managerial AI integration has a negative but statistically insignificant coefficient ($\beta = -0.08$, $p > 0.05$). Therefore, the model does not provide evidence of a statistically significant relationship between managerial AI integration and capital expenditure intensity. Model 3 does not support Hypothesis 1.

Among the control variables in Model 3, firm size has a positive and marginally significant coefficient ($\beta = 0.02$, $p < 0.10$), while firm age has a negative and statistically significant coefficient ($\beta = -0.01$, $p < 0.01$). Leverage is not statistically significant. The adjusted R^2 of Model 3 is 0.03, indicating that the model explains 3% of the variation in capital expenditure intensity. The overall model is not statistically significant.

Overall, the results do not support Hypothesis 1. Managerial AI integration does not have a positive and statistically significant association with any of the three operational indicators. In the labor productivity model, the coefficient is statistically significant but negative. In the ROA

and capital expenditure intensity models, the coefficients are negative and statistically insignificant. Therefore, Hypothesis 1 is rejected.

Table 3. Regression Results for the Direct Effect of Managerial AI Integration

Variable	Model1 Log labor productivity	Model2 ROA	Model3 Capex_intensity
Main effects			
Managerial AI integration	-0.91** (0.327)	-0.09 (0.195)	-0.08 (0.084)
Controls			
Firm size (log assets)	0.18*** (0.032)	-0.06*** (0.018)	0.02 (0.008)
Firm age	-0.03*** (0.011)	-0.01 (0.006)	-0.01** (0.003)
Leverage	0.18 (0.353)	-0.24 (0.188)	0.04 (0.081)
Year effects	Yes	Yes	Yes
Model fit			
Adjusted R^2	0.21	0.11	0.03
F(df)	4.89** (10, 138)	2.92** (10, 138)	1.53 (10, 138)

*Note. Robust standard errors of the HC1 type are reported in parentheses. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.*

Hypothesis 2: The Moderating Role of Firm Size

Hypothesis 2 states that firm size positively moderates the relationship between managerial AI integration and the operational indicators of large Russian IT companies, so that this relationship becomes stronger for larger firms. To test this hypothesis, three regression models with an interaction term were estimated separately for log labor productivity, ROA, and capital expenditure intensity. The key explanatory variables are centered managerial AI integration, centered firm size, and the interaction between managerial AI integration and firm size. Firm age and leverage were included as control variables, and year effects were added to all specifications.

Before interpreting the moderation models, multicollinearity was assessed using the Variance Inflation Factor. The results are reported in Appendix A, Table A2. All VIF values are low: 1.44 for managerial AI integration, 1.31 for firm size, 1.10 for firm age, 1.08 for leverage, and 1.24 for the interaction term. Since all values are below the conventional threshold of 5, the inclusion of the interaction term does not create severe multicollinearity.

The results are presented in Table 4. In Model 1, where log labor productivity is used as the dependent variable, managerial AI integration has a negative and statistically significant coefficient ($\beta = -0.752$, $p < 0.05$). Firm size has a positive and statistically significant coefficient ($\beta = 0.178$, $p < 0.001$). However, the interaction term between managerial AI integration and firm size is negative and statistically insignificant ($\beta = -0.122$, $p > 0.05$). Therefore, firm size does not significantly moderate the relationship between managerial AI integration and labor productivity. Model 1 does not support Hypothesis 2.

In Model 2, where ROA is used as the dependent variable, managerial AI integration has a negative and statistically significant coefficient ($\beta = -0.200$, $p < 0.05$). Firm size also has a negative and statistically significant coefficient ($\beta = -0.053$, $p < 0.001$). At the same time, the interaction term between managerial AI integration and firm size is positive and statistically significant ($\beta = 0.110$, $p < 0.01$). This result indicates that firm size positively moderates the relationship between managerial AI integration and ROA. Therefore, Model 2 supports Hypothesis 2 for ROA.

In Model 3, where capital expenditure intensity is used as the dependent variable, managerial AI integration has a negative but statistically insignificant coefficient ($\beta = -0.006$, $p > 0.05$). Firm size has a positive and statistically significant coefficient ($\beta = 0.016$, $p < 0.001$). However, the interaction term between managerial AI integration and firm size is negative and statistically insignificant ($\beta = -0.026$, $p > 0.05$). Therefore, firm size does not significantly moderate the relationship between managerial AI integration and capital expenditure intensity. Model 3 does not support Hypothesis 2.

Hypothesis 2 is partially supported. The interaction between managerial AI integration and firm size is positive and statistically significant only in the ROA model ($\beta = 0.110$, $p < 0.01$). This indicates that larger firms are better able to transform managerial AI integration into profitability-related outcomes. However, the interaction term is not statistically significant for labor productivity or capital expenditure intensity. Therefore, the moderating effect of firm size is outcome-specific rather than universal.

Table 4. Moderation Analysis: Managerial AI Integration × Firm size

Variable	Model1: Labor productivity	Model 2: ROA	Model3: Capex intensity
Main effects			
Managerial AI integration	-0.264	-0.018	-0.01 (0.035)
Firm size (log assets)	0.18*** (0.053)	-0.05*** (0.012)	0.02*** (0.004)
Interaction			
Managerial AI integration × Firm size	-0.12 (0.119)	0.11** (0.039)	-0.03(0.014)
Controls			
Firm age	-0.00045	-0.000055	-0.002(0.001)
Leverage	0.10 (0.262)	-0.11 (0.091)	-0.0033
Year effects	Yes	Yes	Yes
Constant	2.66*** (0.406)	0.29** (0.112)	0.29*** (0.052)
Model fit			
Adjusted R^2	0.21	0.33	0.09
F(df)	4.55*** (11, 137)	7.58*** (11, 137)	2.31* (11, 137)

*Note. Robust standard errors of the HC1 type are reported in parentheses. *** p < 0.001; ** p < 0.01; * p < 0.05.*

Overall, Hypothesis 1 is rejected, as managerial AI integration is not positively and significantly associated with any of the three operational indicators. Hypothesis 2 receives partial support, as the interaction between managerial AI integration and firm size is positive and statistically significant only in the ROA model.

5.Discussion

This study examined whether managerial AI integration is associated with the operational indicators of large Russian IT companies and whether this relationship depends on firm size. The findings support a conditional rather than universal view of AI value creation. Managerial AI integration was not positively associated with labor productivity, ROA, or capital expenditure intensity. At the same time, firm size positively moderated the relationship between managerial AI integration and ROA, but not the relationships with labor productivity or capital expenditure intensity. Thus, the operational value of managerial AI integration appears to depend on organizational conditions and to vary across performance dimensions.

5.1. Interpretation of the direct effect

The results show that managerial AI integration does not have a direct positive association with the selected operational indicators. In the labor productivity model, the coefficient of managerial AI integration was statistically significant but negative. In the ROA and capital expenditure intensity models, the coefficients were negative and statistically insignificant. Thus, Hypothesis 1 was not supported. This means that the presence of AI in managerial and organizational communication does not automatically translate into higher productivity, profitability, or investment intensity.

The negative coefficient in the labor productivity model may reflect short-term adjustment costs associated with AI integration. The use of AI in managerial activities may require employee reskilling, changes in reporting systems, data preparation, and coordination between managerial and technical functions. These processes can temporarily increase organizational complexity and reduce measured productivity before potential benefits become visible. Therefore, the negative association should be interpreted cautiously as evidence of non-immediate productivity effects rather than as evidence that AI is inherently harmful to firms.

This interpretation is consistent with the resource-based view. From this perspective, technologies do not generate performance advantages by themselves. Mikalef and Gupta (2021) argue that AI capability depends on the firm's ability to select, orchestrate, and leverage AI-specific resources. Therefore, the absence of a positive direct effect does not necessarily mean that AI is irrelevant for operational performance. Rather, it indicates that managerial AI integration alone may be insufficient to generate measurable operational improvements.

The result is also consistent with dynamic capabilities theory. AI may support sensing, seizing, and reconfiguring activities, but the transformation of AI-generated insights into performance outcomes requires changes in routines, resource allocation, and decision-making processes. Mikalef et al. (2020) show that analytics-related capabilities influence performance indirectly through dynamic and operational capabilities. Wamba-Taguimdje et al. (2020) similarly emphasize the role of process-oriented dynamic capabilities in translating AI capabilities into organizational performance. In this study, managerial AI integration may therefore represent an initial condition for performance improvement, but not a sufficient factor by itself.

The period analyzed in this study, 2019–2025, was also characterized by substantial external shocks and restructuring of the Russian IT market. These conditions may have weakened or delayed the observable relationship between managerial AI integration and operational outcomes. This interpretation is relevant to the Russian context because Molodchik et al. (2024) show that the relationship between ICT combinations and financial performance in Russian companies varies across industries and depends on specific configurations of technologies and resources.

The rejection of Hypothesis 1 is also consistent with prior research on implementation barriers. Tariq et al. (2021) identify cultural constraints, fear of the unknown, lack of employee skills, and insufficient strategic planning as barriers to achieving operational excellence through AI. Cannas et al. (2024) show that AI implementation in operations and supply chain management may be constrained by data quality problems, lack of AI-specific skills, high investments, unclear economic benefits, and limited experience in cost analysis. Firms may disclose AI-related managerial initiatives before these initiatives become sufficiently embedded in internal processes to affect productivity or financial outcomes.

5.2. Interpretation of the moderating effect of firm size

The second main finding is that firm size positively moderates the relationship between managerial AI integration and ROA. This means that the association between managerial AI integration and profitability-related operational performance becomes more favorable in larger firms. At the same time, this moderation effect was not found for labor productivity or capital expenditure intensity. Therefore, Hypothesis 2 was partially supported.

This result supports the contingency logic of the study. Contingency theory argues that the effectiveness of managerial practices depends on their fit with organizational conditions (Woodward, 1965). In this case, firm size appears to be an important condition for transforming managerial AI integration into profitability-related outcomes. Larger firms may have stronger

financial resources, broader data assets, more developed digital infrastructure, and more formalized managerial systems. These characteristics may help them use AI-generated insights more effectively in managerial decision-making and resource allocation. This interpretation is consistent with Babina et al. (2024), who show that AI-related growth is concentrated among larger firms. It is also consistent with Ardito et al. (2025), who demonstrate that AI adoption generates stronger benefits when combined with complementary digital technologies such as IoT and big data analytics.

The fact that the moderation effect is significant only for ROA is important. ROA reflects the efficiency of asset utilization and is closer to financial and managerial outcomes than labor productivity or capital expenditure intensity. Larger firms may be better positioned to use AI-supported managerial decisions to improve asset allocation, cost control, and profitability. However, this does not necessarily lead to immediate increases in labor productivity or capital investment intensity. Labor productivity may depend on workforce structure, revenue dynamics, automation effects, and external demand conditions. Capital expenditure intensity may reflect investment cycles and strategic decisions that are not directly determined by AI-related managerial practices.

Therefore, the partial support for Hypothesis 2 suggests that firm size matters, but its role is outcome-specific. The results do not indicate that larger firms benefit from managerial AI integration across all operational indicators. Instead, the evidence shows that firm size strengthens the relationship between managerial AI integration and ROA only. This finding supports the treatment of operational performance as a multidimensional construct rather than a single aggregate outcome.

5.3. Relationship with prior literature

The findings qualify rather than directly contradict prior research on AI and firm performance. Mikalef and Gupta (2021) link AI capability with firm performance, Dubey et al. (2020) connect BDA-AI with operational performance, Babina et al. (2024) show AI-related firm growth, and Ardito et al. (2025) find a positive relationship between AI adoption and revenue growth. However, these effects are usually explained through complementary resources, organizational capabilities, digital maturity, or implementation depth. The present study supports this conditional logic by showing that managerial AI integration alone is not sufficient to generate positive operational outcomes.

This difference can be explained by three factors. First, this study focuses specifically on managerial AI integration, not general AI adoption or AI capability. Managerial AI integration captures the extent to which AI is linked to decision-making, planning, monitoring, and organizational coordination in public corporate communication. Second, the study examines large Russian IT companies during a highly volatile period, when external shocks may have influenced operational outcomes independently of AI use. Third, the dependent variables capture different dimensions of operational performance, and the results show that these dimensions respond differently.

At the same time, the findings are consistent with research emphasizing complementarity in AI value creation. Mikalef and Gupta (2021) argue that AI capability is built through tangible, human, and intangible resources. Mikalef et al. (2020) show that analytics capabilities may affect performance indirectly through dynamic and operational capabilities. The significant moderation effect for ROA supports this perspective by showing that firm size can strengthen the relationship between managerial AI integration and profitability-related outcomes.

The Russian context also makes the findings theoretically relevant. Molodchik et al. (2024) show that ICT profiles in Russian companies are heterogeneous and that their relationship with financial performance varies across industries. This supports the interpretation that AI-related technologies should not be analyzed as isolated tools. Their value depends on broader technological combinations, resource complementarity, and industry-specific conditions. The present study extends this logic to managerial AI integration in large Russian IT companies.

6.Conclusion

This study investigated the relationship between managerial AI integration and the operational indicators of large Russian IT companies. The analysis was based on panel data for 23 companies over the period 2019–2025. The research focused on three operational indicators: labor productivity, ROA, and capital expenditure intensity. It also examined whether firm size moderates the relationship between managerial AI integration and these operational outcomes.

The empirical results show that managerial AI integration does not have a positive direct association with the selected operational indicators. Hypothesis 1 was therefore not supported. In the labor productivity model, the relationship was statistically significant but negative, while in the ROA and capital expenditure intensity models the coefficients were negative and statistically insignificant. These findings suggest that managerial AI integration does not automatically improve operational performance. Instead, AI-related managerial practices may require time, complementary resources, organizational routines, and implementation depth before measurable performance effects appear.

The results also show that firm size positively moderates the relationship between managerial AI integration and ROA. However, this moderating effect was not found for labor productivity or capital expenditure intensity. Therefore, Hypothesis 2 was partially supported. This means that larger firms may be better positioned to transform managerial AI integration into profitability-related outcomes, but this advantage does not extend equally to all operational indicators. The findings answer the research question by showing that the effect of managerial AI integration on operational indicators is conditional, outcome-specific, and dependent on organizational context.

The theoretical contribution of this study is threefold. First, it shifts attention from general AI adoption to managerial AI integration, thereby focusing on the role of AI in decision-making, planning, monitoring, and organizational coordination. Second, it contributes to the literature on AI and operational performance by showing that different operational indicators may respond differently to AI-related managerial practices. Third, it extends the contingency-based view of AI value creation by demonstrating that firm size acts as a boundary condition, but only for profitability-related performance. Overall, the study supports the view that AI creates value not as an isolated technology, but as part of a broader system of organizational resources, managerial routines, and complementary capabilities.

The findings also have practical implications for managers of large Russian IT companies. Companies should not expect managerial AI integration to automatically improve operational

indicators. To generate measurable outcomes, AI-related managerial practices should be supported by data infrastructure, employee skills, clear decision-making processes, and integration into concrete managerial functions such as planning, forecasting, monitoring, and resource allocation. The results also suggest that larger firms may have better conditions for using AI in ways that improve ROA, but size alone is not sufficient. Managers still need to ensure that AI-generated insights are connected to actual managerial decisions and operational processes.

Several limitations should be acknowledged. First, managerial AI integration was measured using publicly available corporate disclosures, which may not fully reflect the actual internal use of AI by managers. Second, the study focused on 23 large Russian IT companies, so the findings may not be generalizable to smaller firms, non-IT industries, or other institutional contexts. Third, the analysis was based on observational panel data; therefore, the results should be interpreted as associations rather than causal effects. Fourth, the period from 2019 to 2025 included major external shocks, including the COVID-19 pandemic, sanctions, and restructuring of the Russian IT market, which may have affected firm-level outcomes.

Future research could address these limitations by using larger samples, longer observation periods, and more direct measures of managerial AI use. Survey data, interviews, or internal company information could help distinguish symbolic AI communication from actual AI-supported decision-making. Future studies could also examine additional moderators, such as digital maturity, data infrastructure, organizational culture, or managerial capabilities. Comparative research across industries or countries could further clarify whether the conditional relationship observed in this study applies beyond large Russian IT companies.

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Appendix

Appendix A

Table A1. Variance Inflation Factor results for Hypothesis 1 models

Variable	VIF
managerial_ai_integration	1.21
Firm size (log assets)	1.31
firm_age	1.10
leverage	1.03

Table A2. Variance Inflation Factor results for Hypothesis 2 models

Variable	VIF
managerial_ai_integration	1.44
Firm size (log assets)	1.31
firm_age	1.10
leverage	1.08
Managerial AI integration × firm size	1.24